G3: Rendering

SFU CMPT 361
Introduction to Visual Computing

In this lecture

- Rendering as a simulation of the “physics of light”
- Basics of rendering: lights, surfaces, cameras & images
- Two families of rendering algorithms

What is **rendering**?

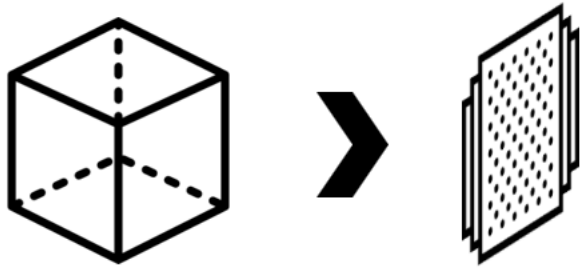
Recall: main subfields in graphics

Main subfields in graphics

Modeling
Computational representations of geometry to create 3D models

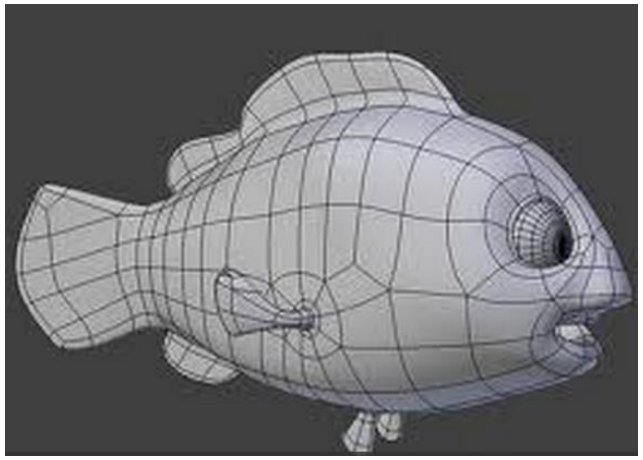
Rendering
Algorithms to generate images from 3D models

Animation
Representing and editing motion with 3D models



Rendering an image

Input: 3D model

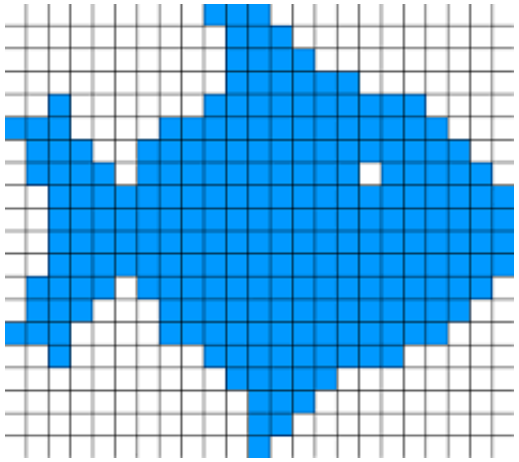


Select viewpoint

+

“render”

Output: 2D image



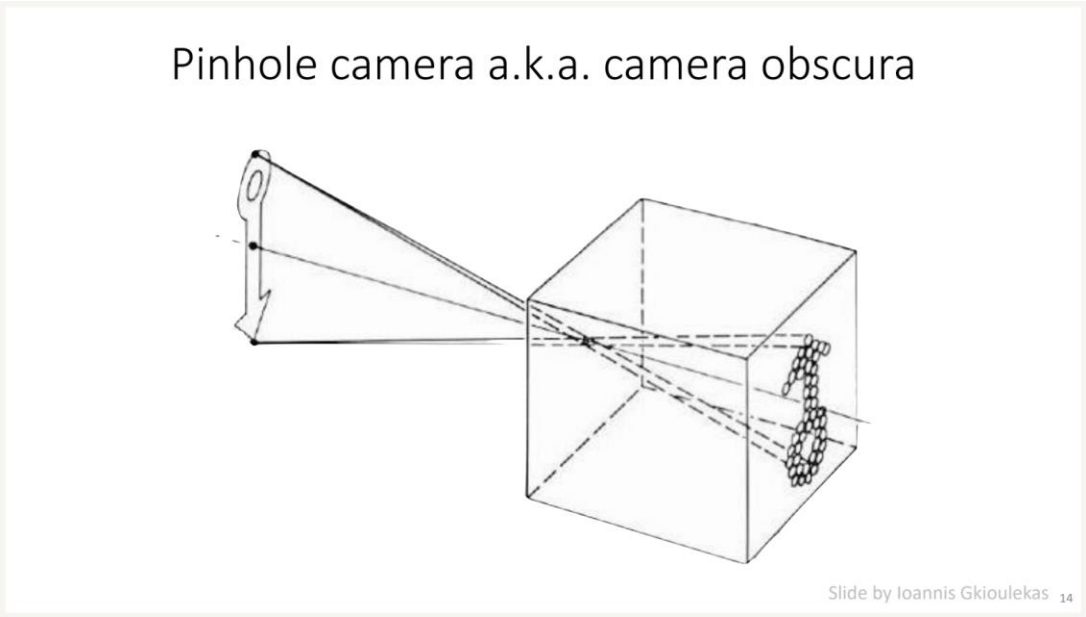
Rendering =
simulating the “physics of light”

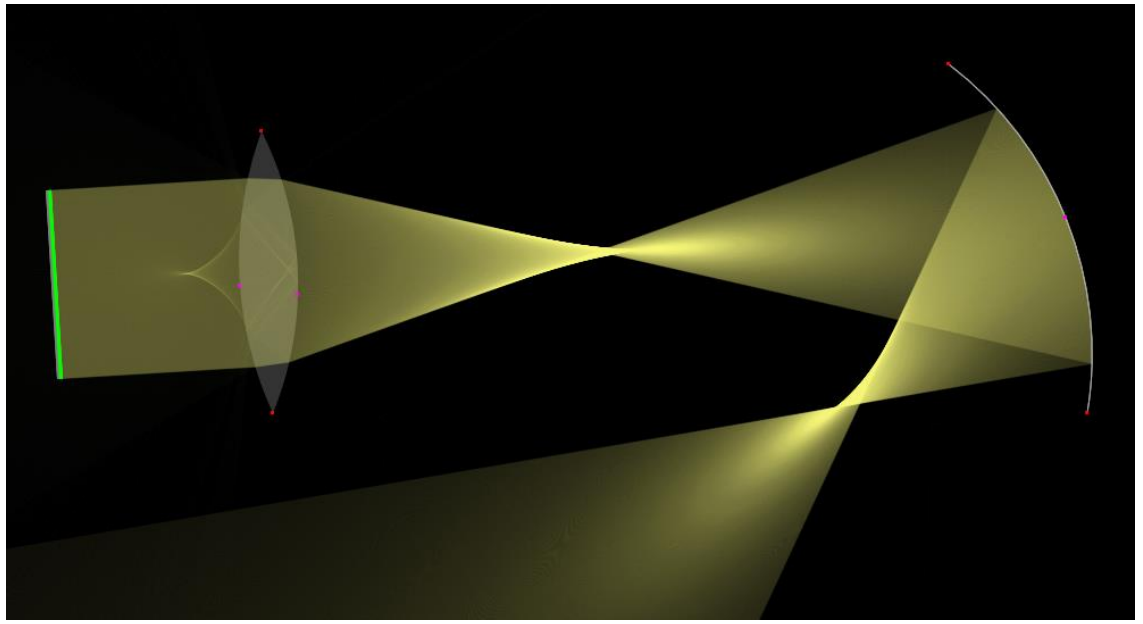
Rendering = “physics of light”

- **Goal**: approximate how things look in the real world
- Light source: **emits** “photons”
- Surface geometry: **reflects** “photons”
- Camera sensor: **receives** “photons”



Simplification: ray optics (geometrical optics)

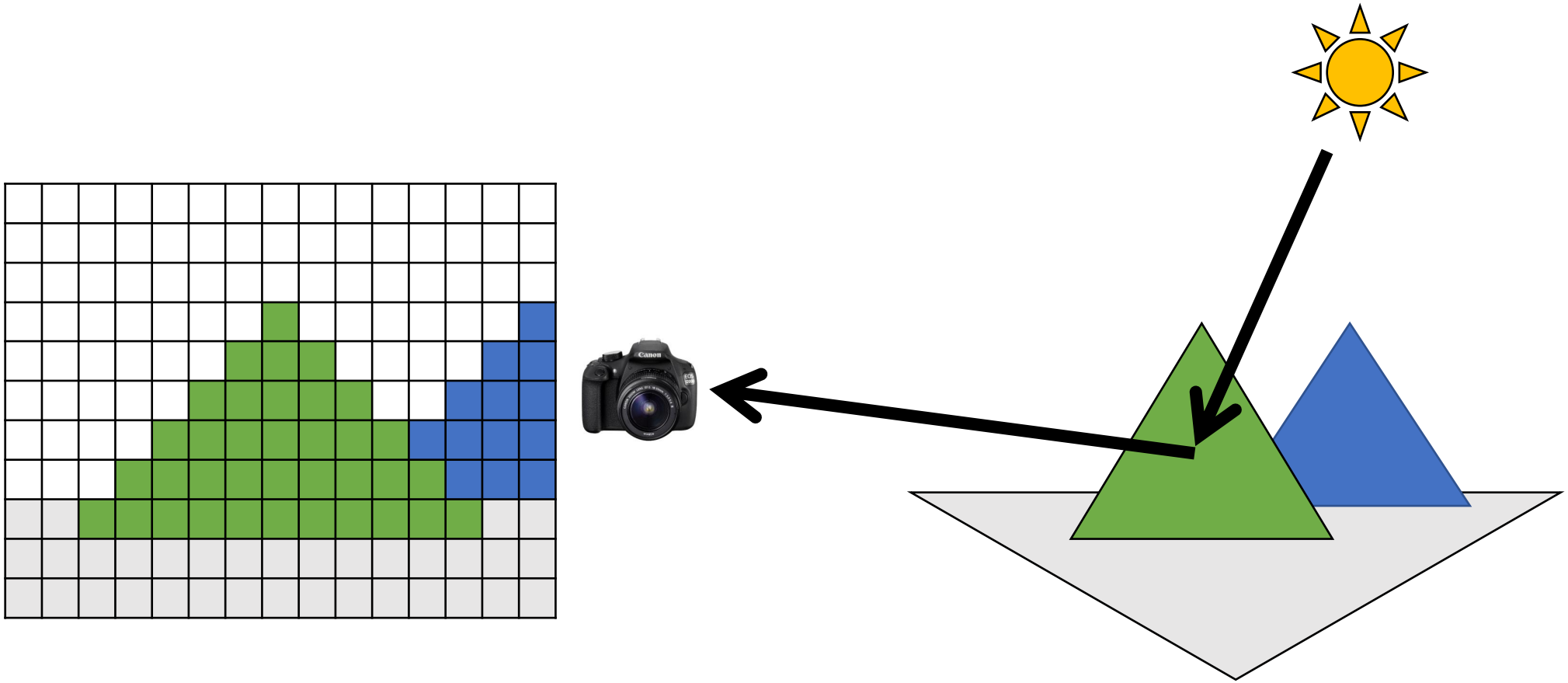




Try out the interactive ray optics simulator by Rick Tu!
https://ricktu288.github.io/ray-optics/

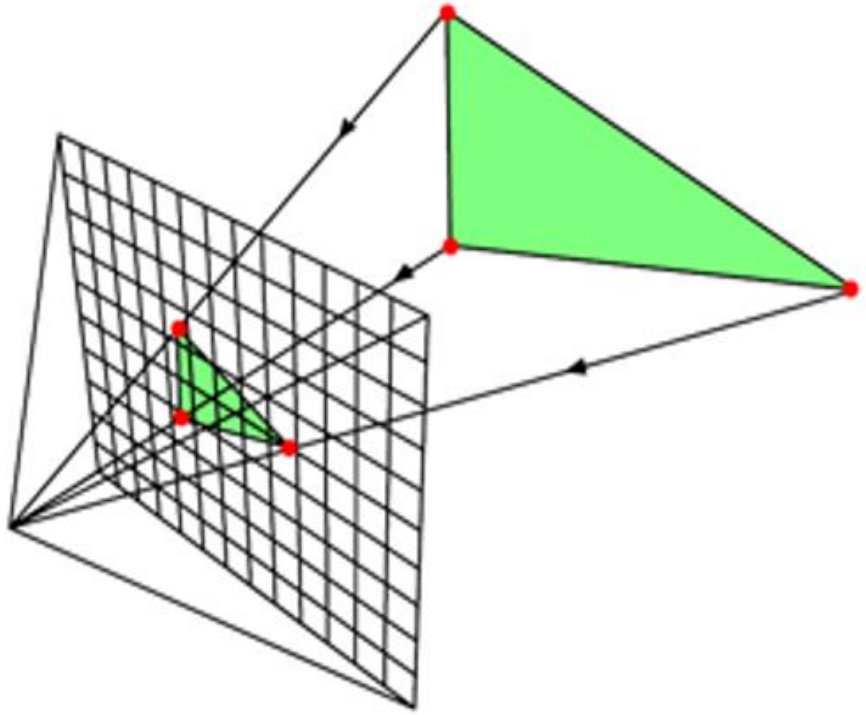
Rendering algorithms: two families

The difference: how we order computation to determine visibility



Rasterization

For each triangle: *“Which pixels are covered by this triangle?”*

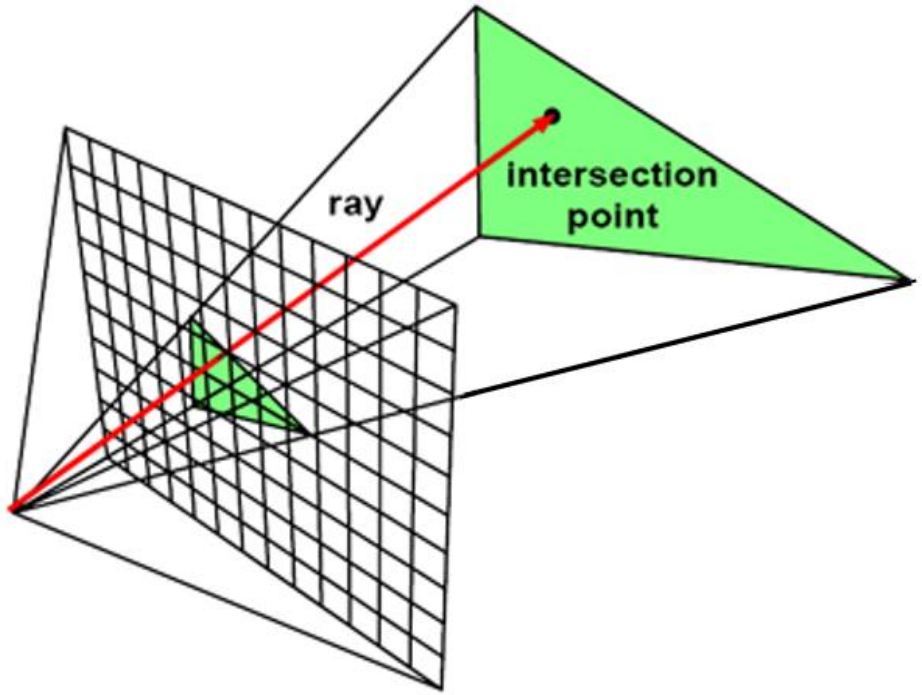


www.scratchapixel.com

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Ray tracing

For each ray (pixel): *“Which triangle is visible from this pixel?”*



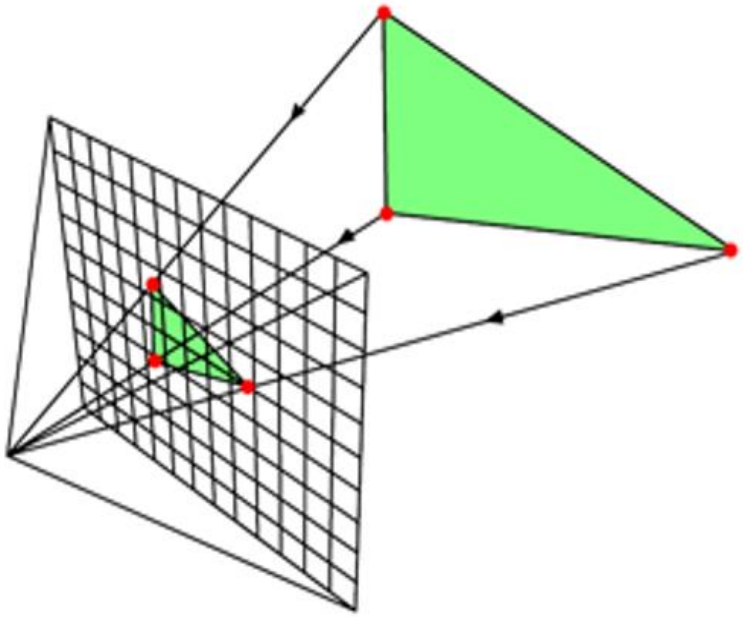
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Rendering Algorithms

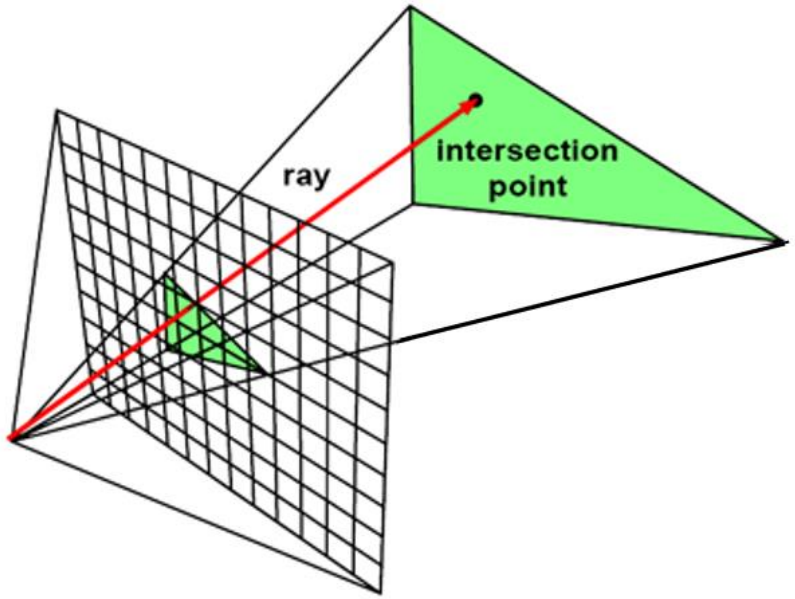
Rasterization

```
For each object {
  // find pixels that are
     covered by the object
}
```



Ray Tracing

```
For each pixel {
  // find objects that are
     visible to the pixel
}
```



Summary

- Rendering as a simulation of the “physics of light”
- Basics of rendering: lights, surfaces, cameras
- Two families of rendering algorithms

References & external resources

- Angel & Shreiner Chapter 8.1-8.2
- Ray Optics Simulation [Rick Tu]
- Physically Based Rendering Book [Pharr, Jakob, Humphreys]
  Donald Knuth: *"This book has deservedly won an Academy Award. I believe it should also be nominated for a Pulitzer Prize."*